15 robots + charging case

Sphero Bolt+ Power Pack

WHAT'S IN THE KIT

- 15 Sphero Bolt+ robots with LCD screens
- Charging case
- Floor tape
- Compasses

QUICK-START GUIDE

1. Same Sphero Edu app workflow as the original Bolt.
2. The LCD screen can show images, animations, and live sensor data.
3. Aim before driving — screen faces you at start.
4. Use the screen for team identity: each squad designs its bot's 'face.'

FULL LESSONS IN THIS GUIDE

***PRO TIP:** Functionally identical to Bolt in the app — mix packs for a 30-robot mega-event.*

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LESSON · GRADES 3-5

Emotive Robot Theater

LEARNING OUTCOMES

- Students will combine movement, screen, and sound into a performance
- Students will sequence a multi-part program
- Students will collaborate on a creative production

MATERIALS

- Sphero Bolt+ robots + Sphero Edu
- Script planning sheet (scene, moves, faces, sounds)
- Simple props and taped 'stage'
- Audience seating arrangement

INSTRUCTIONS

1. Demo the LCD screen: faces, images, and animations on command.
2. Teams write a 30-60 second robot 'play': entrances, moves, emotions on screen.
3. Storyboard first: each beat gets a move + a face + a sound.
4. Program scene by scene, testing each beat before chaining.
5. Dress rehearsal with one partner team as audience.
6. Robot theater: performances with team narration.

ACTIVITY

> Emotive Robot Theater — scripted robot plays where the screen carries the character's feelings. Sequencing a performance beat-by-beat is storyboarding, programming, and drama in one.

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LESSON · GRADES 6-8

Live Data Dashboards

LEARNING OUTCOMES

- Students will display live sensor data on an embedded screen
- Students will design an experiment using a robot as instrument
- Students will interpret real-time data during trials

MATERIALS

- Sphero Bolt+ robots + Sphero Edu
- Experiment menu: speed on carpet vs. tile, ramp angles, collision forces
- Data recording sheets
- Ramps and varied floor surfaces

INSTRUCTIONS

1. Build the instrument: program the LCD to show live speed or heading.
2. Teams choose an experiment from the menu and write a quick plan.
3. Run trials reading the robot's own screen as the data source.
4. Record, average, and chart results across conditions.
5. Swap experiments with another team and attempt to replicate their findings.
6. Discuss: did replication match? Why might it not?

ACTIVITY

> Live Data Dashboards — the robot becomes its own lab instrument, screen streaming live readings during rolling experiments. The replication swap adds real scientific method stakes.

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LESSON · GRADES 9-12

Swarm Choreography

LEARNING OUTCOMES

- Students will coordinate timing across multiple independent programs
- Students will choreograph synchronized multi-robot behavior
- Students will manage a complex technical production

MATERIALS

- Sphero Bolt+ robots (3-5 per group)
- Sphero Edu on multiple devices
- Choreography grid sheet (time vs. robot)
- Music source and taped stage

INSTRUCTIONS

1. The core problem: separate robots, separate programs, one synchronized routine.
2. Groups storyboard on the choreography grid: what each robot does at each timestamp.
3. Program independently, then attempt a synchronized start.
4. Debug drift: robots desync over time — teams engineer their timing fixes.
5. Add screen choreography: coordinated faces and colors across the fleet.
6. Showcase: routines with music, judged on synchronization and ambition.

ACTIVITY

Swarm Choreography — synchronized multi-robot routines where timing drift is the enemy. Managing parallel programs that must agree is a genuine distributed-systems apprenticeship.